

Chapter One

FOURIER SERIES

1.3 Bessel's Inequality and Parseval's Identity

(a) Bessel's Inequality

$$\text{If } f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$
$$2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx$$

(b) Parseval's Identity

$$\text{If } f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$
$$2a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) = \frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 dx$$

Note: Similarly, formula can be written for the limit $-L$ to L

Example: Sawtooth Wave

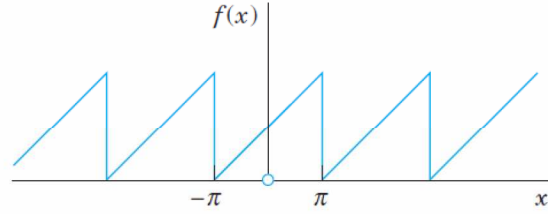


Fig: The function $f(x)$ Sawtooth wave

(a) Find the Fourier series of the function

$$f(x) = x + \pi \quad -\pi < x < \pi \quad \text{and} \quad f(x + 2\pi) = f(x)$$

(b) Prove that $1 + \frac{1}{9} + \frac{1}{25} + \dots = \frac{\pi^2}{8}$

Solution: (a) Let us break the function in to two parts,

$$f(x) = f_1 + f_2 \quad f_1 = x \text{ and } f_2 = \pi$$

The Fourier coefficients of f_2 are zero, except for the first one (the constant term or DC term), which is π . Also, since f_1 is odd the coefficient $a_n = 0$.

$$b_n = \frac{2}{\pi} \int_0^\pi f_1(x) \sin nx \, dx = \frac{2}{\pi} \int_0^\pi x \sin nx \, dx$$

Integrating by parts (uv method) we obtain,

$$b_n = \frac{2}{\pi} \left[x \left(\frac{-\cos nx}{n} \right) - 1 \left(\frac{-\sin nx}{n^2} \right) \right]_0^\pi = -\frac{2}{n} \cos n\pi = -\frac{2}{n} (-1)^n$$

Hence $b_1 = 2, b_2 = -2/2, b_3 = -2/3, b_4 = -4, b_5 = -2/5, \dots$

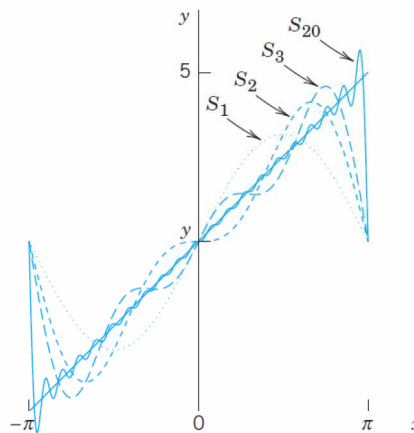


Fig: Partial sums S_1, S_2, S_3, S_{20}

(b)

$$f(x) = \pi + 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right)$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$2a_0^2 + \sum_{n=1}^{\infty} b_n^2 = \frac{1}{\pi} \int_{-\pi}^{\pi} [x + \pi]^2 dx = \frac{8\pi^2}{3}, a_0 = \pi$$

$$1 + \frac{1}{4} + \frac{1}{9} + \dots = \frac{\pi^2}{6}$$